Bidirectional Layout-Semantic-Pixel Joint Decoupling and Embedding Network for Remote Sensing Colorization

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***Abstract*— In recent years, there has been a growing demand for the colorization of remote sensing images due to their inherent limitations caused by remote sensors, such as hazy or noisy atmospheric conditions. These factors result in the captured images needing to be clarified. Compared to ordinary images, remote sensing images present unique challenges in color recovery due to their imbalanced spatial distribution of objects. In this article, we propose a novel bidirectional layout-semantic-pixel joint decoupling and embedding network (BDEnet) following the idea of human painting to generate highly saturated color images with strong spatial consistency and object salience. The proposed BDEnet model emulates the process of human painting through a step-by-step approach. It begins by determining the overall tone of a large macroscopic region and progressively refining the local color based on this initial assessment. Specifically, BDEnet incorporates finer-grained semantics and pixel color information into a colored layout that represents a wide range of continuous areas, thereby accomplishing the colorization task. The BDEnet model operates at three scales, namely the layout (macro), semantic (medium), and pixel (micro) scales. It comprises three key modules: the multiscale feature decoupling (MFD) module, the layout-semantic-pixel multigranularity learning (MGL) module, and the semantic-pixel embedding (SPE) module. MFD module effectively reduces redundant noise from the semantic and layout scales by employing scale decoupling. This process ensures the extraction of efficient features essential for MGL. In the MGL module, three branches with different scales are employed to achieve layout division, semantic segmentation, and pixel coloring. To address the issue of insufficient category label guidance in layouts, we propose a novel approach called similar semantic merging (SSM) using a weakly supervised scheme to accomplish layout division. Finally, the SPE module incorporates stable semantic and pixel information into the layout features. This integration results in the generation of color images that exhibit strong spatial consistency, emphasize object salience, and possess high color saturation.**

***Index Terms*— Colorization, decoupling, multiscale, remote sensing.**

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I. Introduction

NOWADAYS, remote sensing data are becoming increasingly vital in the scientific understanding of Earth processes. The natural pseudocolor image, in addition to multibanded images, is regarded as a crucial way of comprehending morphology following human perception. However, because of the limitations of distance sensors and the air environment, such as haze or clouds, satellite images suffer enormously from either low contrast or chromaticity unsaturation problems. Therefore, color restoration, which is shown in Fig. 1, is eagerly needed in remote sensing images to improve human comprehension and uncover undiscovered Earth processes [1].

With the development of deep learning approaches, image colorization techniques have been extensively researched and made significant progress in approaching human perception. Recently, deep generative models, especially generative adversarial networks (GANs) and their numerous variants [2], [3], [4], [5] are well-researched approaches with satisfactory coloring performance on the ordinary image, which force the generator to produce realistic images through a game with the discriminator.

For image colorization, PatchGAN was proposed by Isola et al. [6], which can produce coloring images with rich chroma. However, from the perspective of space structure, remote sensing images are extremely distinct from regular images since remote sensing images have a contextual structure consistent with the physical Earth surface in which object distribution is quite imbalanced. For instance, there are always macrocontinuous texture areas surrounding microobjects (like vehicles or boats). In this situation, focusing on learning the distribution of local pixels will aggravate the ill-posed problem in macrotexture areas, causing abrupt pixels to be generated in the consistency space. Therefore, the coloring consistency of remote sensing images cannot be guaranteed by the single-scale PatchGAN since it ignores the geographical context.

According to the characteristics of remote sensing images, multiscale remote sensing image colorization methods have been widely studied to ensure spatial consistency. Li et al. [7] introduced a GAN with multiscale discriminators to realize remote sensing colorization. It only optimized the measurement of Jensen–Shannon divergence [8] but failed to fully use

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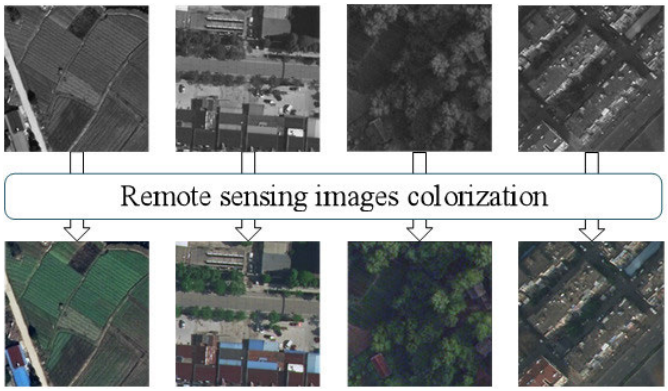


Fig. 1. Examples of remote sensing image colorization. The four top images are grayscale remote sensing images, and the corresponding color image below is color images generated by the colorization approaches.

the macroscale to enhance the spatial stability of the coloring results. Wu et al. [9] proposed an approach using squeeze-and-excitation net (SEnet) to enrich contextual information while assigning various weights to channels, ensuring spatial consistency to a certain degree. Feng et al. [10] proposed a feature extraction method using multiscale residual blocks for the coloring task, and the method carried out information transfer among the extracted multiscale features to realize fusion for generating the color images. Wang et al. [11] proposed joint stream deep convolutional generative adversarial networks (DCGANs) to color the remote sensing grayscales, which achieves the fusion of macroscale and microscale through a gating mechanism. In addition, Wang et al. [12] proposed a bidirectional macro–micro adaptive enhancement network (BMMAEnet) that transfers detailed information from microscale to macroscale to supplement the lost detail at the start of the network and then uses the stable macrohigh-level feature to constrain the micronetwork to generate the color images.

Although the above method can ensure the consistency of remote sensing image color space to a certain extent, it still cannot achieve the optimal effect. Since they leverage different scales to carry out the tasks with the same granularity, that is colorization. In addition, the obtained multiscale features in the above method have the scale entanglement problem, which makes it impossible to mine effective information. This is mainly reflected in the acquisition method of multiscale features, such as using multiple convolution kernels of different sizes or downsampling operations. What's more, the naive fusion method of multiscale features designed by the existing method, such as using addition or dot multiplication operation, cannot make full use of the intrinsic attributes belonging to different scale features.

To generate color remote sensing image with strong spatial consistency and object salience, we propose a multiscale bidirectional layout-semantic-pixel joint decoupling and embedding network (BDEnet) where each scale is used to implement tasks of different granularity. BDEnet is designed to obtain different scale disentanglement features, including layout, semantic, and pixel features, as well as fusing multiscales by the layout and semantic features working as the prior on colorization. Specifically, the BDEnet consists of the multiscale feature decoupling (MFD), multigranularity learning (MGL), and semantic-pixel embedding (SPE) modules. The MFD is utilized to obtain the semantic and layout scale features without redundant noise by decoupling. It can solve the problem of insufficient extraction of the intrinsic attributes of each scale caused by the scale entangled. The MGL module is based on three scales of U-Net networks for multigranularity tasks, which are layout division, semantic segmentation, and pixel coloring. In particular, we propose a similar semantic merging (SSM) scheme based on a weakly supervised method by calculating the second-order correlation between adjacent semantic regions to implement layout division, which addresses the challenge of lacking layout labels. Finally, we propose the SPE module to embed the stable and outstanding semantic and pixel color information in proper order on the layout color feature to generate highly saturated color images with strong spatial consistency and object salience. The main contribution can be concluded as follows.

1) We propose a BDEnet to achieve coloring by simulating the human painting process, which first determines the layout color of a large-scale continuous area, and then gradually embeds semantics and pixel colors on the basis of the layout color. This progressive refinement colorization can ensure the consistency of the coloring space as well as the object salience.
2) We propose the MFD module by decoupling operation to ensure each scale executes the corresponding granularity task. We also propose the SPE module that embeds the pixel and semantic features into the layout to realize fusion adaptively.
3) We validate the effectiveness of the proposed method on the Potsdam dataset and the LoveDA dataset. Several comparison and ablation experiments are performed to demonstrate the effectiveness of the proposed multigranularity decoupling and fusion framework in remote sensing colorizaiton tasks.

II. Related Work

According to whether external information is required, image colorization methods are usually divided into two categories [13], [14], [15]. One is user-based colorization methods, which need to be colored according to auxiliary information provided by the outside, such as scribbles [16], [17], examples [18], [19], [20], etc. The other is the automatic colorization method, which does not need the intervention of external information and only extracts the characteristics of the grayscale image to achieve color restoration.

A. User-Based Colorization Methods

As one of the typical user-based coloring methods, the Scribbling [21], [22], [23] method is diffused into the appropriate range for complete grayscale colorization sing user-provided color scribbles representing a local region as a clue. Sangkloy et al. [21] proposed an adversarial scheme based on GAN to achieve grayscale image coloring which set sketch with local color strokes as the input of the network and the generator output color images by the U-net structure with residual blocks. These methods require a large number of user participation in the process of coloring which is not adapted

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to coloring images with complex texture structures, such as remote sensing images. In addition, choosing the reason color palette can also be challenging.

The Exemplar-based methods [24], [25], [26] utilize the relevant color image as a reference to realize the grayscale colorization. He et al. [25] converted a grayscale picture to a color image, which belongs to the example-based coloring method, by employing a similarity subnet for evaluating the semantic similarity between the provided examples and the grayscale and a colorization subnet with the U-net structure for final colorization by selecting the color and the transfer method according to the similarity. Compared with the scribble-based approaches, the method based on examples requires less reference provided by users. However, their need for high-quality examples is still urgent.

B. Automatic Colorization Method

In recent years, fully automatic grayscale image colorization without additional information has been researched by scholars and compared with user-based methods, it is more suitable for remote sensing images with uneven spatial distribution and rich texture.

Isola et al. [6] proposed an image coloring approach based on a conditional generation adversarial network (CGAN) [3], in which the generator is designed as a U-Net structure composed of the encoder, decoder, and jump connection. In addition, a discriminator is proposed to determine whether the whole image is real or not by dividing the picture into multiple patches and then judging each patch and taking the mean value. It can focus on local areas and optimize the coloring details. Nazeri et al. [27] applied the coloring approach in [6] to the coloring of high-resolution images and proposed a new loss function to optimize the network by maximizing the probability of the discriminator's judgment error which solves the problem of slow convergence in the network training process and improve the stability. Wang and Patel [28] proposed a method, which first processed the noise in remote sensing images that affected the coloring results before the image was input into the generator of GAN. In [6], [27], and [28], the U-Net structure used by the generator loses detail in the encoding process, which is crucial for the coloring task of restoring each pixel value. Therefore, Yun et al. [29] proposed a generator to use the convolution operation with stride 1 for color restoration, which fully guaranteed the coloring details. Although Yun et al. [29] ensures the details of colorization, it ignores the spatial consistency of colorization and leads to discontinuous coloring, which is fatal to remote sensing images.

To alleviate the problem of inconsistent coloring space caused by focusing only on local details, the method of fusing global features has been researched. Iizuka et al. [30] proposed a convolutional neural networks (CNNs)-based colorization method, which fuses the global high-dimensional features extracted in the classification task with the embedded features in the colorization task and then performs color restoration to ensure color space consistency. Similarly, ChromaGAN proposed by Vitoria et al. [31] used the global information obtained from the classification task for coloring through an adversarial training method. However, these methods embed extremely macroscopic semantics for classification into the network, which will cause semantic confusion and low color saturation in the generated color images. Therefore, Zhao et al. [32] proposed saliency map-guided colorization with GAN (SCGAN) to avoid the confusion of different semantic target objects in generated color images through the assistance of the saliency map. However, the saliency map is still not specific enough.

In addition to the use of global features, multiscale method is widely used in remote sensing image colorization to ensure coloring space consistency. Li et al. [7] proposed a GAN with multiscale discriminators, which are used to supervise the features of different scales in each layer in the generator to reduce the ill-posed problem in the coloring process to ensure the consistency of the remote sensing image coloring space. However, this method essentially only optimizes the Jensen–Shannon divergence metric [8] and does not essentially solve the problem of inconsistent space generated in the generation process. Limmer and Lensch [33] proposed a multiscale method in which the initial features of multiple scales were obtained through the feature pyramid, and then the features of different scales were extracted, respectively, which are finally fused also with the detailed information from the shallow layer to ensure the color details and the continual color space. Wu et al. [9] proposed a multi-SEnet model structure to realize remote sensing colorization. They extracted multiscale features using convolution kernels of different sizes and the extracted features were assigned an appropriate weight to each channel through the squeeze-and-excitation scheme to highlight important information. Subsequently, based on multi-SEnet, Wu et al. [34] further transformed the colorization task from RGB to YUV color space to improve the color effect with multiscale convolution to optimize the remote sensing colorization effect. Similarly, Yun et al. [35] proposed multiscale convolutional downsampling and attention mechanism to color the grayscale remote sensing images. The multiscale representation of the above three methods is achieved using convolution kernels of different sizes. Feng et al. [36] combined multiscale feature extraction with the residual network to mine the potential information in the image and used U-Net based on the attention mechanism to generate color images. Based on this work, Feng et al. [10] further used the multiscale residual receptive field to extract features. They also proposed information recovery architecture (IRA) to transfer the down-sampled features to the original scale features to correct the inappropriate information in the original features which can recover the primary color reasonably.

In the above method, macroconstraint and microscale are used to achieve remote sensing image coloring, which has a certain effect on ensuring spatial consistency. However, the way these methods obtain macroscale features is too simple, such as pooling, and convolution kernels of different sizes, resulting in macroscale features being constrained by a fixed receptive field which cannot be a reasonable constraint to the microscale.

Different from the multiscale methods, Zheng et al. [37] proposed a semantic priori coloring method to ensure the

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remote sensing coloring spatial spacial consistency. They calculated the weight of each semantic region in all images and assigned it to the corresponding region and then fit the color feature and semantic weight matrix to realize colorization. Although the strong constraint introduced by Zheng et al. ensures spatial consistency, the salience of the color object is seriously damaged, so it cannot generate a full color image. Therefore, Wang et al. [12] proposed BMMAEnet including microscale, medium scale, and macroscale to generate detail-rich color remote sensing images with high spatial consistency. In the initial stage of feature extraction, BMMAEnet transfers the microscale information to the macroscale to prevent the loss of details in the downsampling process. In the stage of fusion of extracted features, an adaptive macroenhancement module is proposed to enhance the semantic constraints on the microscale. In addition, Wang et al. [11] proposed a joint stream DCGAN consisting of two scale generation networks, which adaptively selected features through a gating mechanism to fuse effective features of two scales to colorize remote sensing images. Wang et al. [11] transferred the shallow details of the macroscale to the deep details of the microscale to supplement the missing details and ensure the object salience.

Compared with the above methods, our proposed BDEnet is a multiscale and multiauxiliary task guidance method, including macroscale, medium scale, and microscale which, respectively, are used for the layout division, semantic segmentation, and pixel coloring tasks belonging to different granularity. BDEnet performs scale decoupling through MSD at the initial stage of feature extraction to remove redundant noise contained in the medium scale and macroscale. The decoupled features are input into the MGL module, which consists of the three scales U-net by different granularity tasks supervision, to obtain the layout division, semantic segmentation, and pixel coloring feature. Finally, based on the most global features containing macrolayout representation obtained during the process of performing the layout division task, the semantic feature obtained from the semantic segmentation task and the pixel color feature obtained from the coloring task is gradually embedded to generate the high color saturation image with strong object salience as well as the space consistency.

III. Method

To ensure coloring spatial consistency as well as object saliency, we proposed BDEnet, which contains layout, semantic, and pixel multigranularity task supervision branches to achieve coloring object salience using pixel scales while ensuring the consistency of the coloring space using layout and semantic scales. We will elaborate on the framework of the BDEnet in the following.

The BDEnet is based on DCGAN [4], including the generator and the discriminator. The generator contains three scales to process tasks with different granularity and includes three major components: the MFD, the MGL, and the SPE module. The input of BDEnet is the original grayscale image I_o, which is set as the input of the layout scale. We downsample the input image to half and a quarter of its original size, expressed as I_h and I_q, through an average pooling operation. I_h and I_q are set as the input of semantic and pixel scale, respectively. First, preliminary feature extraction is performed on I_o, I_h, and I_q by the traditional residual block to generate the initial pixel, semantic, and layout scale feature F_c, F_s, and F_l. F_c and F_s are set as inputs to MFD ① to obtain the decoupled semantic feature F_{ds}. Similarly, the F_{ds} and F_l are set into the input of MFD ② to generate the decoupled layout feature F_{dl}. Then F_c, F_{ds}, and F_{dl} are input into the MGL module and are carried out feature extraction and recovery operations through U-Net, respectively. F_c can generate pixel features F_c and F_{ds} can obtain semantic features F_{ss} under the constraint of segmentation loss. F_{dl} is extracted semantic and generates layout feature F_{ll} by merging the similar combined semantic areas. Finally, we input F_{ll} and F_{ss} into SPE ① and embed the semantics into the layout features. On the basis of them, pixel features are further embedded to generate the final high-saturation image with strong spatial consistency and object salience. We feed the generated image into the discriminator and optimize the coloring results through the game between it and the generator.

A. MFD Module

The input features of the MGL module are essential for realizing layout division, semantic segmentation, and pixel coloring tasks, which directly affect the network performance. It is suboptimal to directly use features from each scale res-block since it will introduce much noise and make the macroscale extremely vulnerable to interference from small-scale objects, which goes against the spatial consistency of coloring. Therefore, we leverage the proposed MFD module to remove noise to extract the latent effective features of each scale.

The MFD module is fed the feature F_c, F_s, and F_l and achieves decoupling in two steps. The first step is obtaining the decoupled semantic feature $F_{\mathrm{ds}} \in \mathbb{R}^{(H/2)\times(W/2)\times C}$ by removing redundant pixel information from the feature $F_c \in \mathbb{R}^{H\times W\times C}$, as shown in the MFD ① module in Fig. 2. Then, the layout features are further decoupled from the feature F_{ds} through the MFD ② module as shown in Fig. 2. The processing of these two modules is similar. Therefore, we take the MFD ①, shown in Fig. 3, as an example to explain in detail.

First, average pooling and upsampling are performed on F_c and F_s, respectively, to obtain feature F_{cd} and feature F_{su}. Second, the decoupling feature F_{d1} is obtained by subtracting F_{cd} from F_s and the decoupling feature F_{d2} is obtained by subtracting F_c from the feature F_{su}. Finally, the feature F_{d1} and feature F_{d2} are adaptively fused by selecting the essential information to obtain the semantic decoupling feature F_{ds}, which is used for semantic color generation. The purpose of using two different methods for decoupling is to make one of them a supplement to the other to remove redundant information fully. The formulas of this module are expressed as follows:

$$\begin{aligned} F_{d_1} &= F_s - \mathrm{Avg}(F_c) \\ F_{d_2} &= \mathrm{Up}(F_s) - F_c \\ F_{d_s}^{(i,j)} &= \frac{e^{F_{d_1}^{(i,j)}}}{e^{F_{d_1}^{(i,j)}} + e^{F_{d_2}^{(i,j)}}} F_{d_1}^{(i,j)} + \frac{e^{F_{d_2}^{(i,j)}}}{e^{F_{d_1}^{(i,j)}} + e^{F_{d_2}^{(i,j)}}} F_{d_2}^{(i,j)} \end{aligned} \tag{1}$$

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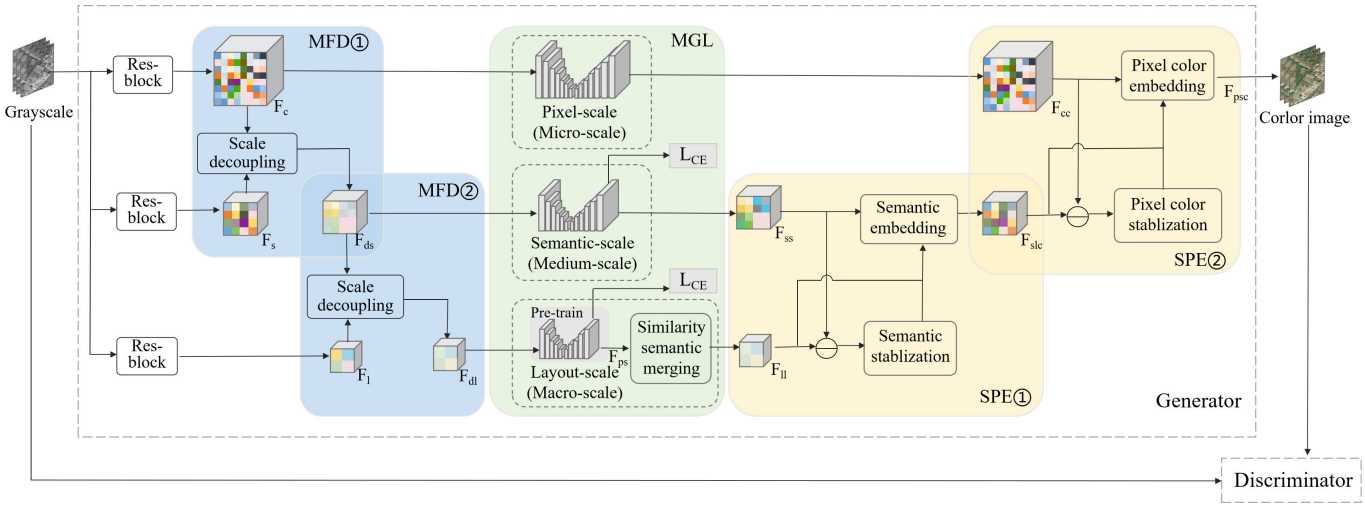


Fig. 2. Overall structure of the BDEnet contains three scales, i.e., layout (macro), semantic (medium), and pixel (micro) including three major components: 1) the MFD module which decouples the entangled features in semantic and layout scales by pixel and semantic feature, respectively, and they correspond to MFD ① and MFD ②; 2) the MGL module extracted the layout, semantic, and pixel color granularity feature in which the semantic, and pixel color feature are obtained by U-net directly while the layout feature obtained by introducing a similarity semantic merging scheme after the macrosemantic extraction by U-net; and 3) the SPE module fuses the features from different scales by gradually embedding the semantic, corresponding SPE ①, and pixel color features, corresponding SPE ②, based on the layout.

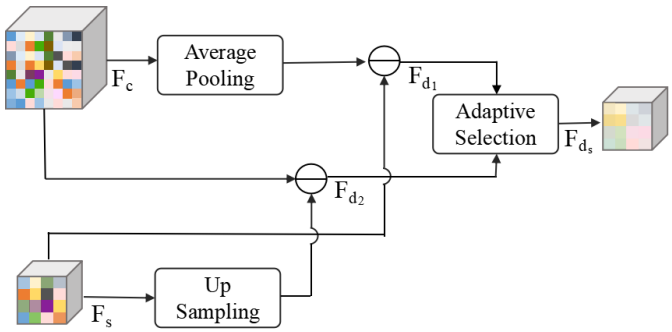


Fig. 3. Illustration of the MFD ① module which utilizes two different processes for feature decoupling and then fuses features obtained from the two by adaptive selection.

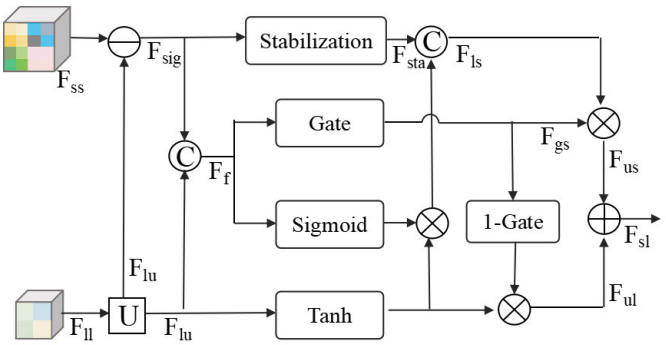


Fig. 4. Illustration of the SPE ① module, which reasonably embeds the significant semantics processed by stabilization in layout feature, which is set as a prior.

where Avg(·) represents the average pooling operation, Up(·) represents the upsampling operation, and i and j are the elements on the ith row and jth column of the corresponding feature.

B. SPE Module

Aiming at generating coloring results with strong spatial consistency as well as object saliency, we proposed the SPE module to fuse the layout, semantic, and pixel coloring features obtained from the MGL module in which the former is more stable while the latter contains more fine-grained coloring information. The SPE gradually refines the color by fusion, including two steps, which are shown in Fig. 4. One is to embed semantic information into layout features to generate more stable semantic-level features with macro-layout constraints, and the other is to embed pixel color features into semantic-level features obtained from step one to generate high-saturation color remote sensing images with salient objects and consistent spaces. It is worth noting that the nonstationary information contained in semantic features and pixel coloring features is easy to lead to an ill-posed problem in the process of colorization, and directly fusing them with layout features will introduce unstable factors and fail to achieve satisfactory coloring results. Therefore, before embedding it into the layout feature (semantic feature), we do a process to strengthen the color expression and smooth out the unstable factors. Since the processing of semantic and pixel color information embedding processes are similar, we take the semantic embedding as an example to explain in detail, corresponding to the SPE ① module in Fig. 2.

First, the semantic feature $F_{ss} \in \mathbb{R}^{H\times W\times C}$ obtained from U-net is subtracted from the upsampled layout feature $F_{lu} \in \mathbb{R}^{H\times W\times C}$ to obtain the significant semantic feature F_{sig}. Then, F_{sig} is stabilized to obtain the feature F_{sta} by the sigmoid function, which erases nonstationary information while strengthening effective information. The operation is shown in the following equation:

$$F_{sta} = \mathrm{Sta}(F_{ss} - \mathrm{up}(F_{ll})) \tag{2}$$

where up() denotes the upsample operation and Sta represents the Sigmoid function.

We fuse the feature F_{lu} and the feature F_{sig} through concatenation to obtain the feature F_f with overall information. Then, we design a gating mechanism to achieve a reasonable embedding fusion of features. First, we obtain the feature F_{ls}

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fusing layout and the stable semantic as shown in the following equation:

$$F_{\text{ls}} = \left(\text{Tanh}(F_{\text{lu}}) \otimes \text{Sigmoid}\left(F_f\right)\right) \| F_{\text{sta}} \tag{3}$$

where "⊗" represent the Hadamard product operation and "‖" denotes the concatenation operation.

Then, the following equation is used to obtain feature F_{us} which contains useful information from the stable semantic information:

$$F_{\text{us}} = \text{Gate}\left(F_f\right) \otimes F_{\text{ls}} \tag{4}$$

where Gate() represents the Sigmoid function.

Similarly, the feature F_{ul} which contains the useful information from the layout information is obtained by the following equation:

$$F_{\text{ul}} = (1 - \text{Gate})\left(F_f\right) \otimes (\text{Tanh}(F_{\text{lu}})). \tag{5}$$

Note that F_{us} and F_{ul} result from adaptive selection, which can ensure a reasonable fusion of semantics and layout information. Finally, fuse F_{us} and F_{ul} to embed essential stable semantic information into the layout to obtain semantic-level features F_{sl}, which is used for constraining coloring.

C. Layout Division Module Based on Similarity Semantic Regions Merging

Different scale networks in MGL modules perform different tasks. Unlike pixels and semantic scales, the supervision information on layout division is difficult to obtain. Therefore, we can not directly use the layout label supervision to achieve layout division. In response to this problem, we propose an SSM scheme, a weak supervision method, to divide the layout. The SSM consists of two parts: calculating the similarity between adjacent semantic regions and merging similar semantic regions.

For the numerous input images, since the semantic regions are unfixed and irregular, calculating the similarity between different semantic regions directly is difficult to achieve. Aiming at this problem, we first divide the semantic features into regular small patches and calculate the similarity between semantics through these small patches representing the semantics to which they belong. Specifically, we divide the features of the last layer of size $h \times w \times c$ extracted from the pretrained semantic segmentation network for semantic segmentation into $h \times w \times n^2$ patches of size $n \times n \times c$ and we set n to 8. Each patch is represented by the centroid of each channel and in size of $1 \times c$. Then, leverage the eight patches around the patch P to represent it and the second-order approach it belongs to can improve the stability of the characterization. Second, similar semantic regions are merged to generate layout partition features. We traverse each patch sequentially and calculate the cosine similarity [38] between two adjacent patches that do not belong to the same semantic region according to their second-order representations and merge the semantic regions whose cosine similarity is less than the threshold. Our proposal for merging similar semantic regions strategy can be summarized as Algorithm 1.

Algorithm 1 Algorithm of Example

Input: Semantic feature F_{ps}
Output: Layout feature F_{ll}
Step 1. Divide feature $F_{ps} \in R^{h \times w \times c}$ into patches $P_i^{(x,y)} \in R^{n \times n \times c}$ of the same size
\\where $i \in (1, (h \times w)/n2)$, (x, y) represents the position
Step 2. Compute the centroid $C_i \in R^{1 \times c}$ of block P_i channel-wise $C_i^z = mean(\sum_{x=1}^{n} \sum_{y=1}^{n} P_i^{c(x,y)}), C_i = (C_i^1, \ldots, C_i^z, \ldots, C_i^c)$
Step 3. Compute the second-order feature representation of the patch P_i
$P_i = \{\}$
for i = 1:$(h \times w)/n^2$ **do**
  for j = 1:8 **do**
    $P_{ij} = C_j - C_i$
    append P_{ij} to P_i
\\ where P_j denote the centroid of patch surrounding P_i
Step 4. Compute semantic similarity
for i = 1:$i - 1$ **do**
  $\cos\theta_{(i,i+1)} = \frac{\sum_{a=1}^{n \times n \times c} (P_i^a \times P_{i+1}^a)}{\sqrt{\sum_{a=1}^{n \times n \times c} (P_i^a)^2} \times \sqrt{\sum_{a=1}^{n * n * c} (P_{i+1}^a)^2}}$
Step 5. Updating labels
if $\cos\theta_{(i,i+1)} > \varepsilon$ **then**
  Set $LabelS_{i+1}$ to $LabelS_i$
\\where S_{i+1} is the semantic region to which patch P_{i+1} belongs, and S_i is the semantic region to which patch P_i belongs
Step 6. Merge similar semantics regions
if $count(LableL) = m$ **then**
  for i = 1:m **do**
    if $S_1, S_2, \ldots, S_n \in L_i$ **then**
      for i = 1:n **do**
        $S_s = S_s + (\sum_{\text{i}=1}^{\text{n}} S_i - S_s)/n$

D. Objective Function

Aiming at improving the similarity between the original color remote sensing image and the generated image and encouraging the generated in line with human perception, we design the objective function to optimize BDEnet. Since the multiscale networks involved in the generation process perform different granularity tasks, optimizing each task is a prerequisite for the overall network to be optimal. Therefore, in addition to GAN losses and Huber losses for the coloring task, our loss function also includes semantic segmentation loss in the layout and semantic scales. The overall loss function is shown in the following equation:

$$\mathcal{L}_{\text{overall}} = \lambda_1(\mathcal{L}_{\text{GAN}} + \alpha\mathcal{L}_{\text{Huber}}) + \lambda_2\mathcal{L}_{\text{seg}} + \lambda_3\mathcal{L}_{\text{lay}} \tag{6}$$

where $\mathcal{L}_{\text{GAN}}$, $\mathcal{L}_{\text{Huber}}$, $\mathcal{L}_{\text{seg}}$, and $\mathcal{L}_{\text{lay}}$ represent the loss of GAN, the loss of pixel color generation, the loss of semantic segmentation, and the loss of layout division, respectively. λ_1, λ_2, and λ_3 are the learnable parameters. We set $\lambda_1 + \lambda_2 + \lambda_3 = 1$ and we set $\alpha = 100$.

1) DCGAN Loss: The BDEnet based on DCGAN includes a generator and a discriminator. The former is used to restore the color information, and the latter is used to judge whether the fed image is actual or generated. In particular, the objective of

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the generator is to make the discriminator unable to distinguish authentic images from generated images, that is, minimize the correct prediction probability of the discriminator, while the objective of the discriminator is to maximize the correct distinguishing probability between the original and the generated image. According to [27], the objective function of the generator and discriminator are shown in the following equations, respectively,

$$\min \mathcal{L}^{(G_i)}\left(\theta_D, \theta_{G_i}\right) = -\mathbb{E}_{x_i P(x_i)}\left[\log(D(G(x_i)))\right] \tag{7}$$

$$\max \mathcal{L}^{(D)}\left(\theta_D, \theta_{G_i}\right) = \mathbb{E}_{y\sim P(y)}\left[\log(D(y|x_i))\right] + \mathbb{E}_{x_i\sim P(x_i)}\left[\log(1 - D(G(x_i|x_i)))\right] \tag{8}$$

where i denotes the ith scale of the generator, and $i = 1, 2, 3$. The optimization of the network is realized by training the generator and discriminator alternately.

2) Color Error Loss: Aiming at generating the authentic color image and encouraging less blurring, previous approaches leverage the traditional loss such as $L1$ distance [6] or $L2$ distance [39] in addition to the GAN objective. However, $L1$ always makes the network difficult to learn due to the large gradient. Meanwhile, $L2$ tends to cause the overall performance degradation of the model due to its large contribution to outliers. Therefore, we choose Huber (or smooth $L1$) [40] as the traditional loss for optimizing the coloring effects. The Huber loss is shown in the following equation:

$$\mathcal{L}_{\text{Huber}} = \begin{cases} \frac{1}{2}(G(x_i) - y_i)^2, & \text{for } |G(x_i) - y_i| < \delta \\ \delta|G(x_i) - y_i| - \frac{1}{2}\delta^2, & \text{otherwise} \end{cases} \tag{9}$$

where, $i = 1, 2, 3$. We set $\delta = 1$ in (11).

3) Semantic Segmentation and Layout Division Loss: The cross-entropy loss [41] is used to supervise the semantic and layout scale task, which is a similarity measurement function. It is worth noting that since the layout division lacks the ground-truth labels, we calculate losses of semantic segmentation networks in the layout scale to optimize layout division indirectly. The loss function of cross entropy is shown in the following equation:

$$\mathcal{L}_{\text{CE}} = \frac{1}{C}\sum_{c=1}^{C}(-y_c \log(p_c) - (1 - y_c)\log(1 - p_c)) \tag{10}$$

where C denotes the number of pixels contained in the image, p_c represents the prediction output of the pixel c by the semantic generation network, and y_c denotes the ground truth label of the pixel c.

IV. Experiments

In this section, we illustrate the dataset, evaluation metrics, and comparison approaches used in the experiment and introduce the vital experimental implementation details. In addition, we validate BDEnet on both qualitative and quantitative metrics by contrast and ablation experiments, and we analyze the reasons according to the experimental data and the visualization effect.

A. Dataset and Implementation Details

Different from previous studies, we choose the International Society for Photogrammetry and Remote Sensing (ISPRS) Potsdam Dataset [42] and the LoveDA Dataset [43] to validate the effectiveness of BDEnet instead of the most common datasets used in coloring tasks, such as NWPU Dataset to avoid the semantic label we require is unavailable.

1) Potsdam Dataset: Potsdam Dataset consists of 38 digital orthophoto remote sensing images from a city with a ground sampling distance of 5 cm. Each image has a size of 6000 × 6000 and contains six semantic categories, namely cars, background/clutter, trees, buildings, impervious surfaces, and low-vegetation areas. The image from the Potsdam Dataset contains four spectral bands, and we choose three bands other than near-infrared, namely red, green, and blue bands for experiments. We divide each entire picture containing 6000 × 6000 pixels into multiple small images in the size of 256 × 256 and randomly select sixty percent of them as the training set and divide the rest equally into the verification set and the test set.

2) LoveDA: LoveDA is a dataset for remote sensing image classification released by the RSIDEA team of the State Key Laboratory of Remote Sensing Information Engineering of Wuhan University. It contains remote sensing images of both urban and rural areas, which contain multiscale objects from Nanjing, Wuhan, and Changzhou. The dataset contains 5987 images that can be divided into seven categories, with 166 768 semantic annotation objects, and each image width is 1024. We divided them into 256 × 256 images to use and set them into a training set, verification set, and test set according to the ratio of 6:2:2.

B. Experiment Details

The BDEnet we proposed is trained on four NVIDIA Tesla V100 GPUs using the PyTorch framework [44]. The network input is a processed 256 × 256 single-channel grayscale map, and the output is an RGB three-channel color map. The batch size is set to 16, and we train 50 epochs to improve the performance of BDEnet using the Adam algorithm [45] with a learning rate of 0.003. The trained model parameters are used for verification and testing, and the best results from testing are taken as the final experimental data of the BDEnet.

The number of BDEnet parameters is 39.83M, the average training time for each epoch is 147 s, and the inference time for a single picture is 0.28 s. To alleviate the problem of a large number of parameters, we use the strategy of parallel computing for training, which reduces time costs and improves efficiency. The traditional method trains each epoch at 488 s, and the parallel computing training strategy reduces the training time per epoch by 341 s.

C. Evaluation Metrics

To verify the effectiveness of the proposed BDEnet, we evaluated its performance through four indicators, namely, peak signal-to-noise ratio (PSNR) [46], structural similarity (SSIM) [47], mean absolute error (MAE) [27] accuracy [27], and Amazon Mechanical Turk (AMT) [6]. Among them, AMT is a qualitative metric and the other three are quantitative.

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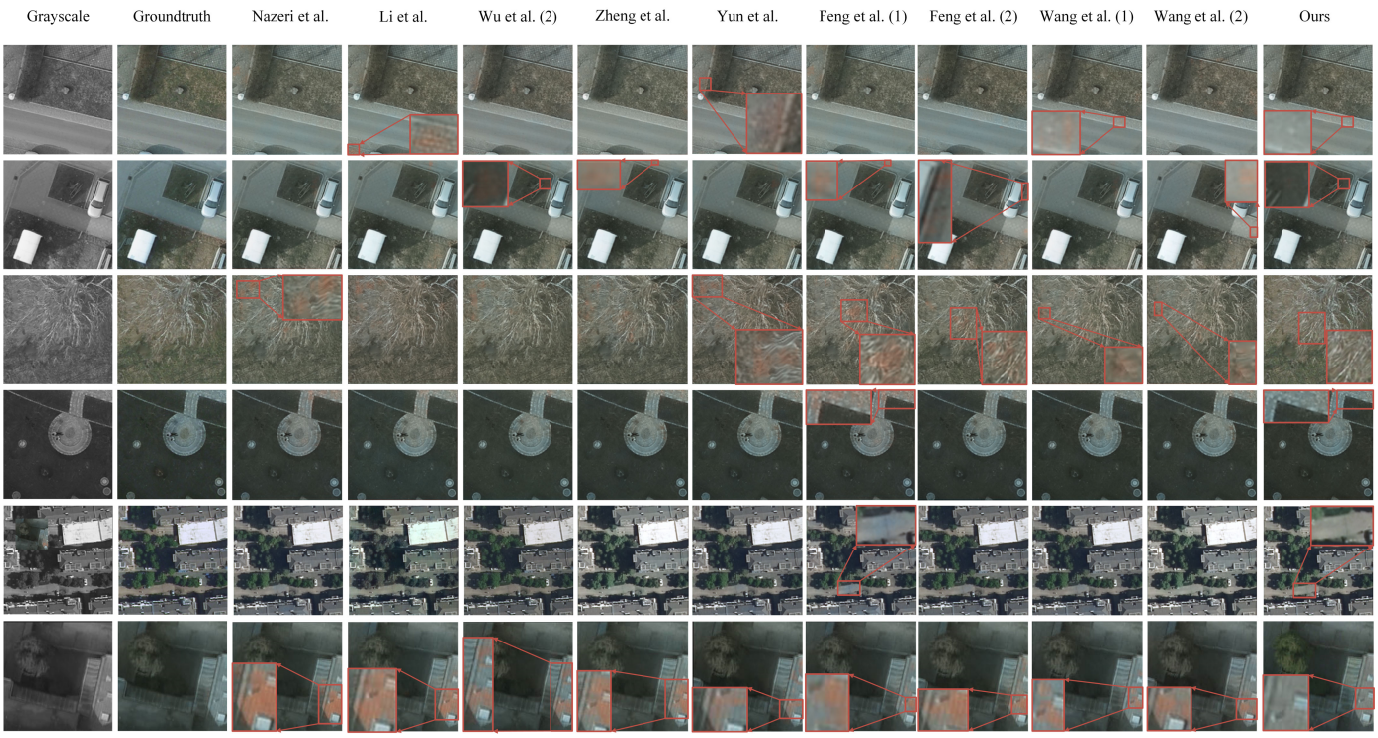


Fig. 5. Several examples of generated images by the contrast methods and BDEnet. By comparing the color images generated by different approaches, it is obvious that the images generated by BDEnet have stronger spatial consistency and object salience and are more in line with human understanding.

TABLE I
CONTRAST EXPERIMENTS RESULTS ON POTSDAM AND LOVEDA DATASET

Method	Potsdam Date Set					LoveDA Data Set				
	PSNR	SSIM	MAE	Acc	AMT	PSNR	SSIM	MAE	Acc	AMT
Isola et al. [6]	25.73	89.83	12.93	87.02	52.50	28.45	89.46	10.65	76.23	42.50
Nazeri et al. [27]	25.96	88.20	12.49	85.43	50.00	28.02	90.56	9.82	77.58	47.50
Li et al. [7]	26.33	90.15	11.07	86.11	52.50	28.49	90.07	10.79	77.35	45.00
Wu et al. (1) [9]	27.16	91.63	10.58	85.98	60.00	28.73	90.57	9.32	78.09	45.00
Wu et al. (2) [34]	27.38	92.45	10.03	86.92	60.00	28.90	90.98	9.01	78.23	47.50
Zheng et al. [37]	26.64	92.92	9.92	87.03	57.50	28.85	90.01	8.46	77.72	45.00
Yun et al. [35]	27.73	94.04	9.11	87.85	65.00	29.73	91.52	8.07	78.47	52.50
Feng et al. (1) [36]	27.90	92.75	9.54	87.14	62.50	28.79	89.24	9.48	78.98	50.00
Feng et al. (2) [10]	28.37	93.24	8.24	87.09	65.00	29.35	91.66	8.51	79.74	50.00
Wang et al. (1) [11]	30.74	94.03	7.41	88.24	67.50	30.08	91.62	7.54	80.35	52.50
Wang et al. (2) [12]	30.97	94.12	7.39	87.93	70.00	29.86	91.96	**7.52**	80.61	55.00
Ours	**32.05**	**96.60**	**6.02**	**89.97**	**75.00**	**32.71**	**94.48**	7.67	**81.91**	**60.00**

1) PSNR [46]: We choose PSNR as one of our quantitative evaluation metrics, which is used commonly to assess the generated image quality, and a higher PSNR usually indicates a higher colorization quality. The formula is as follows:

$$\mathrm{PSNR} = 10 \times \log_{10}\left[\frac{(L)^2}{\mathrm{MSE}}\right], \quad \mathrm{MSE} = \frac{1}{N}\left\|y - \hat{y}\right\|_F^2 \tag{11}$$

where R and N represent the dynamic range of pixel values and the number of pixels, and $\|\cdot\|_F^2$ represents the Frobenius norm.

2) SSIM [47]: SSIM, as a quantitative evaluation metric, measures image similarity in contrast, brightness, and structure. The value of SSIM is between 0 and 1, and the value closer to 1, the higher the quality of the generated image is. The formula is as follows:

$$\mathrm{SSIM} = \frac{\left(2\mu_y\mu_{\hat{y}} + c_1\right)\left(\sigma_{y\hat{y}} + c_2\right)}{\left(\mu_y^2 + \mu_{\hat{y}}^2 + c_1\right)\left(\sigma_y^2 + \sigma_{\hat{y}}^2 + c_2\right)}$$
$$c_1 = (k_1 L)^2, \quad c_2 = (k_2 L)^2 \tag{12}$$

where μ_y, $\mu_{\hat{y}}$, σ_y^2, and $\sigma_{\hat{y}}^2$, respectively, denote the mean of y, the mean of $\hat{y}$, the variance of y, and the variance of $\hat{y}$, as well as, $\sigma_{y\hat{y}}$ denotes the covariance of y and $\hat{y}$. k_1 and k_2 are equal to 0.01 and 0.03, respectively.

3) MAE [27]: MAE is computed by taking the mean of the absolute error of the generated and source images on a pixel level for each color channel.

4) Accuracy [27]: We set the accuracy rate as another quantitative metric to measure the performance of each approach.

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Fig. 6. Examples for validating the object salience. (Left column) Color images generated by BDEnet. (Right) Other contrast method. The area circled using red boxes validates that the generated images by BDEnet have stronger object salience and higher color saturation.

The formula is shown as follows:

$$\mathrm{acc}(x, y) = \frac{1}{n}\sum_{p=1}^{n}\prod_{l=1}^{3} 1_{[0,\varepsilon_l]}\left(\left|h(x)^{(p,l)} - y^{(p,l)}\right|\right). \quad (13)$$

In (13), n represents the total number of positions in the generated image, p represents a position, l denotes the channel, sigma denotes the threshold, and $G(x)$ and y denote the generated image and the real image, respectively. According to (13), accuracy can be explained as the percentage of the total positions that are accurately colored, where accuracy is defined as the pixel difference between the three channels on a position and the real image is less than the threshold simultaneously. Here, we set the threshold to 5.

5) AMT [6]: We applied the AMT perception test to verify the performance of each method and present the experimental data. The qualitative metric of AMT is actually crucial for the task of color, because the ultimate goal of colorization is to generate color images that conform to human senses. In contrast, only the use of the qualitative can not necessarily measure the visual effect of the image.

We recruit 20 people to take part in the AMT test and presented 20 sets of images to the participants successively. Each set of images included grayscale images, authentic color images, and generated color images. Participants are asked to judge which of the two color images is true or false. If the participant is wrong or unable to judge, the image is considered to have been successfully colored. Finally, the mean value of the success rate of the judgment of all participants was calculated as the experimental data of the AMT.

D. Comparative Network

1) Isola et al.'s Method [6]: Use CGAN to realize colorization which sets the initializing random noise as the condition.

2) Nazeri et al.'s Method [27]: Use DCGAN to perform remote sensing image coloring, which sets the original grayscale images as the prior condition.

3) Li et al.'s Method [7]: Use multiscale discriminator GAN to discriminate features from each decoder layer of the generator to achieve remote sensing colorization.

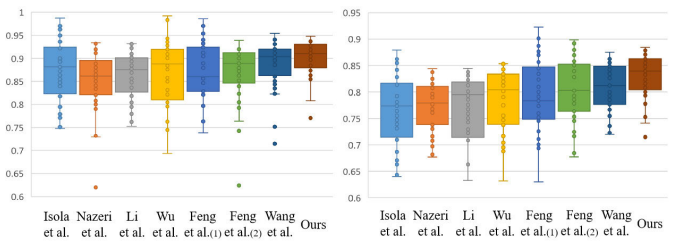


Fig. 7. Box plots of accuracy distributions for the BDEnet and other contrast approaches, which show the BDEnet has higher accuracy metrics and is more stable. (a) Potsdam dataset. (b) LoveDa dataset.

4) Wu et al.'s Method (1) [9]: Use DCGAN composed of multiscale convolution and extrusion-excitation network (SEnet) to achieve remote sensing image colorization.

5) Wu et al.'s Method (2) [34]: Use multiscale DCGAN to color remote sensing images, which are achieved by performing convolution operations with multiple convolution kernels of different sizes in parallel.

6) Zheng et al.'s Method [37]: Use semantic significance prior to guiding the remote sensing image coloring and introducing focal loss to resolve the effects of sample imbalance.

7) Yun et al.'s Method [35]: Extract multiscale features from the encoder and then calculate channel attention, which is added to the corresponding decoding layer to realize the color generation of remote sensing images.

8) Feng et al.'s Method (1) [36]: Use the U-Net as the basic structure and use multiscale residual receptive field blocks (MRRFBs) to realize the extraction of multiscale features.

9) Feng et al.'s Method (2) [10]: Based on the method proposed in [36], the IRA module for information transfer between multiscale features is added to recover the color at the pixel level.

10) Wang et al.'s Method (1) [11]: Use a Joint-stream generator in which the enhanced shallow features of the macroscale are transferred to the deep layer of the microscale and use an adaptive gating mechanism to fuse the multiscale features.

11) Wang et al.'s Method (2) [12]: A multiscale network which preserves macroscale details through a downward decouple module and realizes multiscale fusion through an upward attention module.

E. Contrast Experiments

To verify the performance of BDEnet, we compare BDEnet with other methods proposed in recent years for colorization, especially for the colorization of remote sensing images. Table I shows the experimental results of our method and comparison methods on the Potsdam dataset and LoveDA dataset, and Fig. 5 shows the visualization effects.

We enlarge the regions in Fig. 5 to compare the generated effects more intuitively. In the third and sixth rows, the image generated by methods proposed in [27] has poor spatial consistency. The reason for this phenomenon is that Nazeri et al.'s method uses a single scale in which the limitation of the receptive field cannot make the network capture the

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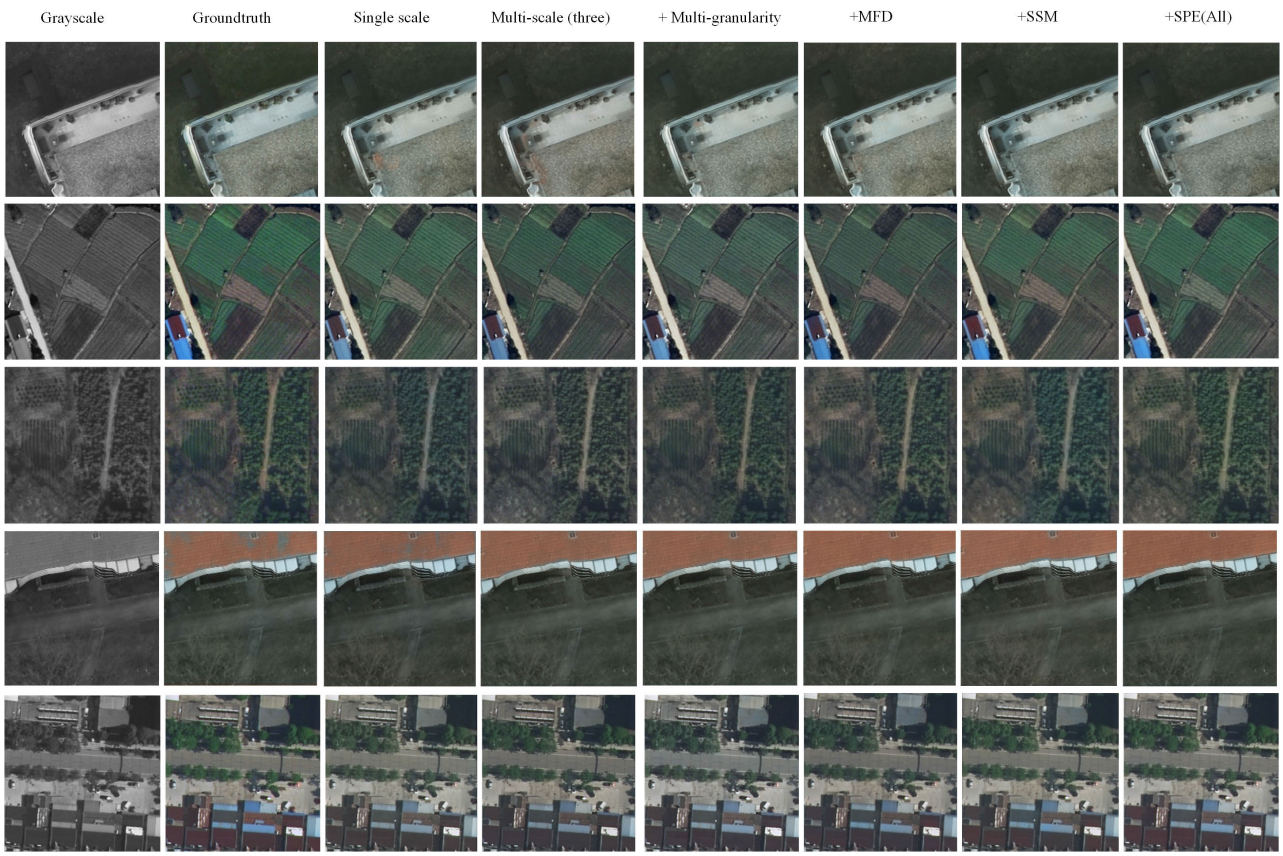


Fig. 8. Generated images of the ablation experiments. "+Multigranularity" represents adding the semantic segmentation loss base on the simple three-scale net to supervise the semantic and layout scales to obtain the stable macrogranularity features. "+MFD" represents adding the MFD module based on the former structure to achieve MFD. "+SSM" denotes adding the process of similarity semantic merging to force the layout scale to learn the more stable layout-grained feature. "+SPE" denotes utilizing the SPE module to realize the multiscale feature fusion by gradually embedding semantic and pixel color features into the layout.

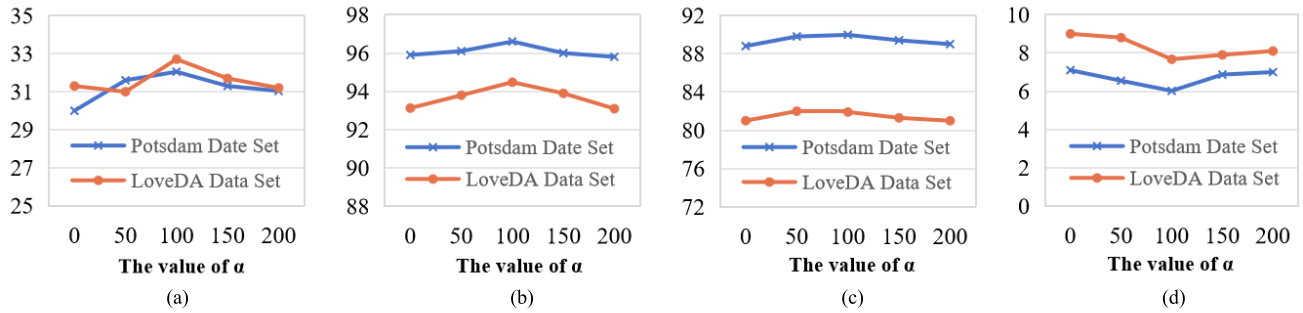


Fig. 9. Performance with different values of the parameter α. (a) PSNR. (b) SSIM. (c) MAE. (d) Acc.



Fig. 10. Visualization effect of the generated features during the MFD module. F_c and F_s denote the original pixel and semantic scales feature and F_{ds} is the decoupled semantic feature obtained by removing the redundant noise in F_s according to F_c. In addition, F_l is the original layout feature and F_{dl} is the decoupled layout feature obtained by removing the redundant noise in F_l according to F_{ds}.

long-distance dependence between pixels, which is fatal for remote sensing images with uneven spatial distribution.

Compared with the above two methods, the spatial consistency of other comparison methods has been enhanced to a

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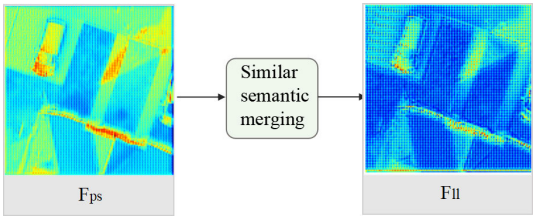


Fig. 11. Visualization effect of the similarity semantic merging. F_{ps} is the last layer feature in the macroscale U-net, and F_{ll} denotes the feature obtained by merging the similarity semantic regions in F_{ps}.

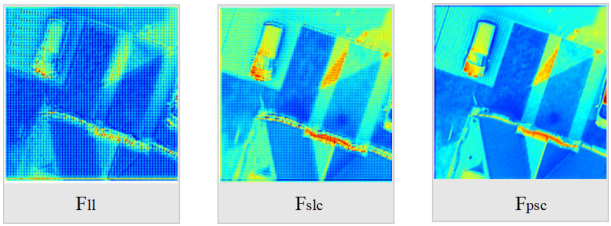


Fig. 12. Visualization effect of the generated features during the multiscale fusion process in the SPE module. The feature F_{ll} denotes the high-level layout feature. F_{slc} represents the feature embedding the semantic in F_{ll}, and F_{psc} represents the feature embedding the pixel color in F_{slc}.

certain extent because they abandon the single scale and use the multiscale network for coloring to extract macrocontext information to ensure the spatial consistency of coloring. However, these multiscale methods are still unable to obtain satisfactory results. In Fig. 5, we can clearly find that in the color image generated by the comparison method, there are obvious patches that are prone to disharmony with the surrounding area. For example, in the sixth row, the roofs generated by the comparison method, even by the BMMAEnet proposed by Wang et al. [12], all appear red patches, breaking the global consistency of coloring. However, in the image generated by our method, the roof area is still a realistic gray roof. This is because the comparison method does not decouple between scales when performing multiscale feature extraction, which leads to numerous redundant noises in the features extracted at different scales. In addition, the methods they used to obtain macrofeatures that constrain microscales, such as pooling and convolution kernel at different scales, are limited by fixed receptive fields. Therefore, the macrofeatures they obtained are not optimal to work as a constraint to ensure coloring space consistency. What's more, the inappropriate fusion method applied in them fails to give full play to the constraint effect of the macroscale as a prior condition on the microscale.

The method proposed by Zheng et al. [37] puts forward the idea of semantic priori, which assigns the same fixed parameters to elements belonging to the same semantic region with the weight of the number proportion of semantic regions. This semantic constraint method is too fixed and cannot be adjusted adaptively so it is difficult to achieve satisfactory results, such as the color images shown in the sixth column of the second and sixth rows.

Compared with the above methods, the color images generated by our proposed BDEnet are continuous and smooth between pixels globally without abnormal light spots appearing. They have strong spatial consistency and are closer to the original color images and more in line with the human senses.

Moreover, Fig. 6 shows a comparison example generated by the suboptimal method proposed by Wang et al. [12] and BDEnet to illustrate the effect of strong object salience. For instance, in the first group of comparison images, the roof has a more apparent red color generated by BDEnet, which can clearly distinguish it from the surrounding environment. In contrast, the roof is almost submerged in the surrounding trees generated by other methods. In the second group of images, the three car objects in the images generated by BDEnet have rich color, high saturation, and strong salience, which is significantly better than the comparison method.

More specifically, the BDEnet further improves PSNR, SSIM, and accuracy by 1.08%, 2.48%, and 2.04% and decreases MAE by 1.37% compared with the suboptimal BMMAEnet [12] on the Potsdam dataset and decreases. In addition, BDEnet increases PSNR, SSIM, and accuracy by 1.85%, 2.85%, and 1.30% than the BMMAEnet [12] on the LoveDA dataset, which proves that our proposed model with applying multiscale decoupling and fusion and multigranularity tasks are entirely practical.

Finally, aiming at verifying the stability of the BDEnet, we select the experimental process data of accuracy metric to draw box plots, as shown in Fig. 7. The median, upper quartile, and lower quartile of the data generated by our method are higher than that of other methods, and the data distribution is more compact. Therefore, box plots indicate that our model has a better effect and more stable performance on both Potsdam and LoveDA Dataset.

F. Performance With Different Values of the Parameter α

The free parameter α in (6) is leveraged to adjust the weight of the Huber loss. According to [6], we vary α from 0 to 200 with intervals of 50 to search for the optimal value of α. The results are illustrated in Fig. 9, and we can obtain the following observation.

As α increases in the range of 0–100, the capability of the model improves steadily. Then, the performance reaches the saturation point when α is 100. This phenomenon illustrates that color error loss plays a considerable role in the remote sensing colorization framework training. After this, the experimental results on all metrics begin to decline. Therefore, we set $\alpha = 100$ for the training process.

G. Verification of the Effectiveness of Multiscale

To verify the effectiveness of our proposed BDEnet, we conducted sufficient ablation experiments, and the experimental results are shown in Table I. The single-scale network proposed by Nazeri et al. [27] is used as the baseline for the experiments. Compared with the single-scale network, the performance of the multiscale network is improved, which is due to the large receptive field at the macroscopic scale being able to extract the dependencies between long-distance pixels, which is crucial for feature extraction of remote sensing images with the uneven spatial distribution.

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(a) (b) (c) (d) (e) (f)

Fig. 13. Visualization of the larger scale color remote sensing images. (a)–(d) In the size of 1024 × 1024. (e) and (f) In the size of 2048 × 2048.

TABLE II
ABLATION EXPERIMENTS RESULTS ON POTSDAM AND LOVEDA DATASET

	Single scale	Multi-sclae (two)	Multi-scale (three)	Multi-granularity	MFD	SSM	SPE	Potsdam				LoveDA			
								PSNR	SSIM	MAE	Acc	PSNR	SSIM	MAE	Acc
①	✓							25.96	88.20	12.49	85.43	28.02	90.56	9.82	77.58
②		✓						26.39	88.33	12.03	86.20	29.69	92.07	10.04	78.25
③			✓					27.50	91.78	10.05	86.89	29.74	92.63	9.46	78.51
④			✓	✓				28.96	93.99	8.57	87.76	29.95	9.29	9.41	79.42
⑤			✓		✓			29.09	93.54	8.24	87.36	30.42	92.97	9.07	79.14
⑥			✓	✓	✓			30.41	94.76	8.33	88.14	30.89	93.59	8.78	79.93
⑦			✓	✓		✓		29.60	94.66	7.93	88.45	30.55	93.25	8.73	79.72
⑧			✓	✓	✓	✓		30.91	95.88	7.32	88.92	31.65	93.90	8.37	81.36
⑨			✓	✓	✓	✓	✓	**32.05**	**96.60**	**6.02**	**89.97**	**32.71**	**94.48**	**7.67**	**81.91**

TABLE III
EXPERIMENTS RESULTS OF DIFFERENT FUSION METHODS ON POTSDAM AND LOVEDA DATASET

Fusion method	Potsdam				LoveDA			
	PSNR	SSIM	MAE	Accuracy	PSNR	SSIM	MAE	Accuracy
Addition	30.91	95.88	7.319	88.92	31.65	93.90	8.37	81.36
Concatenation	30.44	95.14	7.52	89.10	31.28	92.36	8.41	80.14
Selection	31.51	95.75	6.67	89.79	31.97	93.89	7.95	81.87
Gate scheme	31.30	**96.37**	6.83	**90.15**	32.28	93.18	7.63	81.21
Ours	**32.05**	96.20	**6.32**	89.97	**32.71**	**94.48**	**7.67**	**81.91**

H. Verification of the Effectiveness of Multigranularity Task

In the MGL module, we force the macroscale and medium scale to perform the multigranularity task of semantic segmentation and layout division, which are set as a prior to coloring, by introducing a segmentation loss. The experimental results in the third and fourth rows in Table II show that the introduction of the multigranularity task implemented with segmentation loss constraints improves the performance of the model because the features within the object can be clustered by segmentation task, which actually improves the stationarity of the features and eases the ill-problem. In addition, as shown in Fig. 8, compared with the network without the setting of the multigranularity task, it can more precisely assist microscale coloring in the unit of the region where an object is located to improve the spatial consistency of coloring. The effectiveness of the multigranularity task is further verified by comparing the experimental data in the fifth and sixth rows in Table II.

I. Verification of the Effectiveness of the MFD

To verify the effectiveness of the decoupling module, we further introduce the decoupling module for experiments based on the above module. On the Potsdam dataset, the PSNR and SSIM are improved by about 1.6% by adding the decoupling module between the multiscale. This is because the macroscale directly extracts features from the down-sampling grayscale image, resulting in the coupled macroscale features containing information that already exists in the microscale features, which is redundant noise for the macroscale and is not conducive to realizing the macrotask. The proposed MFD module can remove the redundant noise in the macroscale to mine potential features for modeling scale tasks. Fig. 10 shows the visualization effect of the generated features during the decoupling process. Taking MFD ① as an example, we can see that the decoupled semantic features F_{ds} and F_s have less noise, and the features in the same semantic region are more cohesive than the coupled semantic features.

J. Verification of the Effectiveness of the SSM

Comparing the experimental data between the fourth and the seventh row and between the seventh and the eighth row in Table II, it can be concluded that the SSM module is effective. The layout features are obtained by transferring information between different similar semantic regions belonging to the same layout through SSM, which further coarsens the region division grain. The semantic features and pixel color features are embedded in the coarse grain layout feature to further ensure the spatial consistency of the coloring. Fig. 11 shows a visual example of merging.

K. Performance of Different Fusion Methods

To verify the effectiveness of our proposed fusion method, we compared the fusion method used in the other multiscale

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network for remote sensing image coloring with the SPE fusion module. Among them, addition and concatenation methods are naive and they directly aggregate the features from multiscale without any processing, which will drown the valuable information in a large number of useless information. The fusion methods proposed in [11] and [12] improve the utilization of practical information by processing multiscale features and then fusing them through feature selection and gating mechanisms, which have a specific improvement compared with direct fusion. However, the effect of these two fusion methods applied to our proposed BDEnet is still inferior to SPE since the SPE fusion method generates color images by gradually and adaptively embedding stable salience semantic and pixel color features into the continuous macrolayout feature, which ensures coloring space consistency as well as object salience and high color saturation of the generated image. Table III shows the experimental data obtained by different fusion methods. Fig. 12 shows an example map of the fusion process of gradually embedding semantic and pixel color information into the layout.

L. Larger Scale Color Remote Sensing Image Generation

In this work, we divided the image into 256 × 256 small pieces for color to balance the number of parameters and the amount of calculation. Nevertheless, in certain situations, a more global coloring image is required. Therefore, we design a splicing method to obtain the coloring effect of larger size color images. First, we splice the two blocks with the Alpha fusion method to prevent obvious gaps between independent blocks. Then, we use the histogram matching [48] to coordinate the tone between image blocks. Fig. 13 shows the 1024 × 1024 and 2048 × 2048 color images obtained by the splicing method, and we can find that larger images can still have a satisfying effect with spatial consistency and object salience.

V. Conclusion

In this article, we propose a novel BDEnet for coloring remote sensing images with unbalanced spatial distribution. The BDEnet simulates the process of human painting, takes the layout division as the prior knowledge, and gradually embeds semantic and pixel colors based on it to complete the coloring work of the whole image. Specifically, BDEnet is a three-scale DCGAN, including three modules, namely MFD, MGL, and SPE module. MFD is used to remove redundant noise in the macroscale and obtain decoupling features to realize the task for the different scales. Then, the MGL module utilizes decoupling features to achieve multigranularity tasks, including layout division, semantic segmentation, and pixel coloring where the former is used as a prior to the latter to ensure the coloring space consistency. Due to the lack of labels for layout division, a semantic merging mechanism is proposed to achieve the division of layout, which contains the macrocontinual regional. Finally, based on layout features leveraged to guarantee strong spatial consistency, the semantic features and pixel color features processed by the stabilization are gradually embedded to obtain high-saturation color images with object salience further. We verify BDEnet on Potsdam and LoveDA datasets, and the experimental results show that our method is generally superior to the state-of-the-art. The multiscale leveraged in BDEnet increases the computational cost while obtaining superior results. In the future, we will focus on designing lightweight models to reduce computational costs.

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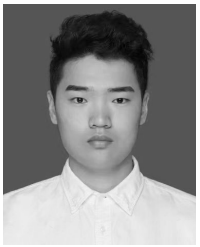
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